Computer Science — Class 6

Single National Curriculum (SNC) — Punjab Curriculum and Textbook Board

Unit 3 — Digital Skills — Simplified Exam Notes for School-Based Assessment (SBA)

TOPIC 1: SOFTWARE AND ITS TYPES:

Software is a set of instructions, data and programs that are used to operate a computer and perform a specific task. There are two basic types of software: system software and application software.

1. **System Software:** System software is the set of useful programs found on every computer that is designed to control and manage the computer's internal and external resources. ***Example:*** the Operating System is the most important system software; it works as the primary interface between the hardware and the user applications.
2. **Application Software:** Application software refers to programs written to carry out a certain task for the user. ***Example:*** Microsoft Word for writing text, Microsoft Excel for spreadsheets, Microsoft Paint for editing images, and Internet Explorer for browsing the internet.

Long Question: Discuss system software. *[Exercise]*

System software serves three core purposes, and it comes in four basic types.

1. **Simplifies Hardware Use:** It simplifies the use and the interaction of the hardware and software.
2. **Aids in Control:** It provides an aid in controlling the hardware and the software of the computer.
3. **Optimizes Performance:** It optimizes the effectiveness and the performance of the computer.
4. **Operating System:** The operating system is the system software that controls and manages all the operations of the computer.
5. **Device Driver:** A device driver is the system software that lets a hardware part communicate with the computer.
6. **Utility Program:** A utility program is the system software that performs a management task related to the computer's functions.
7. **Language Translator:** A language translator is the system software used to translate a computer program from human-readable language into machine-understandable code.

TOPIC 2: OPERATING SYSTEM AND ITS FUNCTIONS: The operating system is a system software that controls and manages all the operations of the computer, and a computer does not function without one.

Long Question: Write down the functions of OS. ***[Exercise]***

The operating system performs the following main functions:

1. **Booting Process:** The OS completes its own loading into the main storage before the computer starts.
2. **Security Management:** The OS manages the security of the user accounts.
3. **Application Management:** The OS enables the user to install and use the application programs on the computer.
4. **Memory Management:** The OS organizes the use of memory and processing time between the user and the programs.
5. **File Management:** The OS enables the user to save, load, delete, rename, print and copy files.
6. **Device Management:** The OS controls the use of peripherals such as disk drivers and printers.
7. **Coordination between Software and User:** The OS provides an interface between the user and the computer's software and hardware.

Some of the most common operating systems are Android, MacOS, MacDOS, Chrome OS, Blackberry, Solaris, Linux and Microsoft Windows (like Windows 10, 8, 7, Vista, XP etc.).

TOPIC 3: TYPES OF OPERATING SYSTEMS: There are several well-known operating systems, and each one is built for a different kind of device.

Long Question: Discuss any two examples of operating systems. ***[Exercise]***

Three widely used operating systems are described below, so that even if one is forgotten, the others are still remembered:

1. **Windows:** Windows is a graphical operating system developed by Microsoft that allows a user to operate the computer. ***Example:*** it is used for personal computers, workstations and embedded systems.
2. **Mac:** Mac OS is the computer operating software used by Apple desktops and laptops. ***Example:*** it was introduced in 1984 for Macintosh personal computers.

3. **Linux:** Linux is a Unix-like, community-developed, open-source operating system for computers, servers, mainframes, mobile devices and embedded devices. ***Example:*** it is supported on almost every major computer platform, including x86, ARM and SPARC.

A fourth operating system worth knowing is Android, which is a mobile operating system that is categorized as open-source software; it is developed by Google, is based upon the Linux Kernel and GNU software, and it targets tablet computers and smartphones.

TOPIC 4: DEVICE DRIVERS, UTILITY PROGRAMS AND LANGUAGE TRANSLATORS: Besides the operating system, a computer relies on three more kinds of system software to keep working smoothly.

1. **Device Drivers:** Device drivers are the system software that enable the computer's hardware components to communicate with the computer. ***Example:*** a printer driver and a sound driver.
2. **Utility Programs:** A utility program, also known as a service program, is an application that performs management tasks related to the functions of the computer. ***Example:*** antivirus, file management tools, compression tools, disk cleanup tool, disk management tools, and backup utility.
3. **Language Translators:** Language translators are software that are used to translate a computer program from human-readable language to machine-understandable code language. ***Example:*** Assembler, Compiler, and Interpreter.

TOPIC 5: APPLICATION SOFTWARE AND ITS TYPES: Application software is software that is developed to aid the user in performing a specific task on the computer, and it comes in several kinds.

Long Question: What is application software? List down its types. ***[Exercise]***

Application software is software that is developed to aid the user to perform a specific task, and it is divided into the following types:

1. **Entertainment Software:** Entertainment software is software that is solely designed for the sake of entertainment. ***Example:*** video games, audio, music, animations and cartoons.
2. **Productivity Software:** Productivity software is software that is used to produce information in the form of a text document. ***Example:*** spreadsheets, presentations, digital art and digital painting.
3. **Education and Reference Software:** Education and reference software is software that is primarily used to enhance and assist learning skills, especially while researching and developing on educational grounds. ***Example:*** encyclopedias, dictionaries, educational games, Google Maps, and tutorials.

TOPIC 6: EXPLORING MICROSOFT WINDOWS: After a Windows computer starts, the desktop is the first and the main screen a user interacts with, and it can be customized and controlled through several built-in tools.

Startup Procedure of Windows: Starting a Windows computer and logging in follows a fixed sequence of steps.

1. **Step 1:** Press the power button available on the system unit; the login screen will appear.
2. **Step 2:** Click the user name and write the password in the password field.
3. **Step 3:** Press the Enter button or click on the login button.

Parts of the Desktop: The desktop is made up of a few main parts that a user sees and uses every time the computer is on.

1. **Icons:** Icons are the small pictures on the desktop that can open a file, folder or program.
2. **Taskbar:** The taskbar is a long bar that is usually found at the bottom of the desktop.
3. **Background:** The background is the large area of the desktop where all the opened programs are displayed.
4. **Start Button:** Clicking the start button opens the start screen, from which the user can open any program found on the computer.
5. **Program Buttons:** The area of the taskbar with several program buttons shows the icons of the programs that are currently running, and clicking an icon opens that program.
6. **Notification Area:** The notification area is found on the right side of the taskbar, and it shows the date and time, along with icons for things such as battery, Wi-Fi, volume, calendar and the action center.
7. **Show Desktop Button:** The show desktop button is found on the far right of the taskbar, and clicking it hides all open windows so the desktop becomes visible.

File Explorer is a file manager application that allows access to files and folders on the computer. The desktop background, also called a wallpaper, can be changed by right-clicking anywhere on the desktop, choosing "Personalize" from the menu, clicking on desktop background, and then selecting an image.

Snipping Tool: Snipping tools are used to take screenshots of a specified area on the screen; on Windows 10 and 11 it can be opened with the keyboard shortcut Windows + Shift + S. **Example:** it offers a Free-form Snip, a Rectangular Snip, a Window Snip, and a Full-Screen Snip.

Microsoft Windows Store: The Microsoft Windows Store is a digital distribution platform that has nearly everything for a user's device, and it provides different types of applications free of cost or on payment.

Working with the Active Window: The active window is the currently used program that is opened in front of all the other windows on the desktop, and it is controlled using three commands on the top right of its title bar.

1. **Close:** This button allows the user to close the active window.
2. **Maximize:** This button allows the window to enlarge and fill the entire desktop screen, or restore it back.
3. **Minimize:** This button hides the active window from view and creates a button for it on the taskbar.

TOPIC 7: MANAGING FILES AND FOLDERS:

The core function of the computer is to sort and organize information and data, and this is done using files, folders and shortcuts.

1. **File:** A file is a common storage unit on the computer. ***Example:*** the stored information can be a text, photo, audio, video, document or presentation.
2. **Folder:** A folder, also called a directory, is a collection of files kept in order to organize them on a storage device. ***Example:*** a folder is a virtual place for documents, applications and data, and it can even hold other folders inside it as sub-folders.
3. **Shortcut:** A shortcut is a link to a file that is created to easily access it. ***Example:*** shortcuts can be created on the desktop and in other folders, and clicking on a shortcut takes the user straight to the original file.

Creating a New File: A new file can be created anywhere on the computer in simple steps.

1. **Step 1:** Navigate to the location or the folder in the computer where the file should be created.
2. **Step 2:** Right click on any empty space in that folder.
3. **Step 3:** A menu will open in place; select "New" from the popped-up menu.
4. **Step 4:** Select the type of file to be created in that place.
5. **Step 5:** Name the newly created file by simply clicking on it.
6. **Step 6:** Once the file is created, double click on its icon to open it.

Creating a New Folder: Creating a folder follows almost the same process as creating a file.

1. **Step 1:** In the location where the folder is needed, select the option to create a folder.
2. **Step 2:** After the new folder appears, click on it to name it.
3. **Step 3:** A file or folder can be dragged into the folder by clicking it and holding it with the help of the mouse.

4. **Step 4:** Double clicking the folder shows the files that it contains.

Create a Shortcut: A shortcut of a file or a folder can be placed on the desktop in two steps: right click on the folder or the file so a menu pops up, then select the option "Create shortcut" from that menu.

Cut, Copy and Paste a File, Folder or Shortcut: Moving or duplicating a file follows three basic steps.

1. **Step 1:** Select the file or folder that needs to be cut or copied.
2. **Step 2:** Right click on the file or folder and select the cut or copy option from the menu that appears.
3. **Step 3:** Right click on the location where the file or folder should be pasted, and select "Paste" from the menu that appears.

Delete and Restore a File or Folder: A file that is no longer needed can be removed, and it can usually be brought back afterwards.

1. **Delete:** Select the file, right click on it, and choose the delete option from the menu; the deleted file or folder is then kept safely in the recycle bin.
2. **Restore:** Restoring means bringing a file, folder or shortcut back to its original place from the recycle bin. ***Example:*** double click the recycle bin icon on the desktop to open it, right click on the file that needs to be restored, and click the "Restore" option to put it back in its original location.

TOPIC 8: WORKING WITH PAINT 3D:

Paint 3D is a raster graphics and 3D creative application, and a refresh of Microsoft Paint, that was released in 2017, and it has numerous 2D and 3D tools that work together to create fun and professional projects.

Paint 3D Interface: The Paint 3D window is made up of a few key working areas.

1. **Title Bar:** The title bar is the strip at the top of the window that shows the name of the open file.
2. **Drawing Tools:** The drawing tools hold the brushes, shapes, text and effects options used to create a picture.
3. **View Tools:** The view tools let the user switch between a 2D view and a 3D view of the canvas.
4. **Brushes:** The brushes panel lists the different brush types that can be selected for drawing.
5. **Selection Tools:** The selection tools, such as Select, Crop and Magic select, are used to choose part of the picture to work on.
6. **Active Tool Panel:** The active tool panel on the right shows the options for whichever tool is currently selected.

Brushes: Hovering the mouse over a brush tool displays its name, and selecting a brush and clicking and dragging from a starting point draws a line with it.

1. **Choosing a Brush:** Hover the mouse over the brush tool to display the name of the brush, then select the brush of choice.
2. **Drawing with It:** Start drawing lines by clicking on the left mouse button and dragging it from the starting point. ***Example:*** there are several brush sizes, colours and shapes to select from.
3. **Saving the Project:** From the Menu, click on the save option, enter the name of the file, and click save in the dialog box that appears.

Creating 2D Shapes: A ready-made 2D shape can be placed and resized on the canvas in a few steps.

1. **Step 1:** Select the 2D shapes option from the toolbar.
2. **Step 2:** After selecting the shape of choice, drag it into the drawing area.
3. **Step 3:** Modify the shape by using the handles by the side of the shape to alter its size and shape.

Making a 3D Image: A flat picture can be turned into a 3D one in three steps: design a 2D image, select the 3D mode to make the image three-dimensional, and then rotate the object and resize it using the handles on the side.

Clone Stamp Tool: The clone stamp tool is used to copy the images on the canvas.

1. **Step 1:** Drag the required shape into the drawing area.
2. **Step 2:** Select the stamp tool, and a new copy of the shape appears.
3. **Step 3:** Repeat this process for as many shapes as are required.

Drawing Straight Lines and Multi-Point Curves: A straight line of any length and direction can be drawn using the line tool.

1. **Step 1:** Select the 2D shapes tool.
2. **Step 2:** Select the straight-line tool.
3. **Step 3:** Drag the mouse from the starting point to stretch a line of the desired length in any direction.
4. **Step 4:** Control the line in any direction by selecting its corners or its midpoints.

TOPIC 9: WORKING WITH 3D MODELS:

A 3D model is a technique in computer graphics that is vastly used to create real-world objects along three axes to represent their height, length and width, giving an illusion of moving in and out of the plane of the canvas.

1. **3D Shapes:** In geometry, 3D shapes are defined as shapes with three dimensions, such as length, width and height.

2. **Create 3D Objects:** Select the 3D shapes tool from the toolbar, select a cube object, drag the cube out and rotate it around each axis, and then change its colour or flip it horizontally or vertically.
3. **Create 3D Doodles:** A doodle is defined as an aimless or random drawing or scribbling. ***Example:*** select the 3D shapes tool, select the 3D doodle option, select a colour, and then drag out from the options while clicking the mouse to create a free-hand shape or curve, which can later be modified by rotating it around its axis.
4. **Create 3D Text:** The text tool provides several options for writing in 2D or 3D and for changing colours, sizes, fonts, styles and alignment. ***Example:*** select the text tool and its style options, select the font and the style options, click outside the text box, and then edit, rotate and flip the text as required.
5. **Use Image Stickers:** Select the 3D shapes tool, select the 3D mode and drag out an object, select the stickers tool, add the eye and mouth stickers to complete the animated look, and confirm the stickers by rotating the object.
6. **The Canvas:** The canvas works like a 2D sheet of paper, and the 3-dimensional objects can be moved in front of or behind the plane of the canvas by selecting the Canvas tool.

TOPIC 10: NAVIGATING THE INTERNET: The internet can be explored using a web browser and a search engine, both of which are examples of application software.

Long Question: Write a note on web browsers. ***[Exercise]***

A web browser is an application software that opens webpages to allow a user to view information from the internet, and moving from one page to another while viewing information is known as surfing the internet.

1. **Examples of Web Browsers:** Internet Explorer, Google Chrome, Opera, Safari and Mozilla Firefox.
2. **Address Bar:** The address of the webpage that a user wants to visit can be typed in the address bar.
3. **Menu Button:** The menu button gives access to useful commands, such as opening a new window, printing, and managing bookmarks.
4. **Webpage Section:** The webpage section displays the content of that webpage.
5. **New Tab Button:** Clicking the new tab button opens a new tab in the browser.
6. **Other Browser Buttons:** A browser window also has a Title Bar, a Refresh button, an optional Home page button, a Favorites Bar, a Profile button, Settings, Back and Forward navigation, and shortcuts from the browsing history.

Search Engine: A search engine is a website that allows a user to search for information on the internet. **Example:** Google, Yahoo, Kidzsearch, Ask and Microsoft Bing.

1. **Researching a Topic:** A search engine can be used to research a topic and gather information about it.
2. **Finding Media:** A search engine can be used to find and save pictures and images.
3. **Downloading Files:** A search engine can be used to search for and download files and programs.

TOPIC 11: USING RAPID TYPING TUTOR:

Rapid Typing Tutor is one of the most used software applications that is used to learn and enhance the keyboarding skills of the user, and keyboarding means typing on the keyboard using the correct finger for each key.

Prescribed Sitting Posture when Keyboarding: A correct sitting posture is important for comfortable and accurate keyboarding.

1. **Back and Elbow:** Sit upright and keep the back straight, and keep the elbow bent at an angle of 90 degrees.
2. **Head and Screen Distance:** Face towards the monitor with the head slightly inclined forward, and keep an ideal distance of at least 45 to 70 cm from the screen.
3. **Shoulders and Wrists:** Sit in a way that the shoulders, arms and wrist muscles are relaxed, and the wrist should touch the table top in front of the keyboard so that the body weight is not shifted onto the wrists.

Home Row Position: The home row position is the specific key on which each finger is required to rest on the keyboard, using both hands with the thumbs resting on the space bar.

1. **Left Hand:** The fingers from the left hand should be placed on the keys A, S, D and F.
2. **Right Hand:** The fingers from the right hand should be placed on the keys J, K, L and the semicolon key.
3. **Returning to Home Row:** When typing, the fingers should always return to this primary placement of ASDF and JKL after pressing another key.
4. **Using the Pinky for Shift:** The Shift key is always pressed with the pinky finger on both hands.
5. **Accuracy over Speed:** Speed is secondary, whereas accuracy is more important while practicing typing at equal intervals.

Mistake Found in the Book: The book states that the keyboard shortcut "Alt + Del" permanently deletes a file without sending it to the recycle bin. The correct Windows shortcut for permanently deleting a file is **"Shift + Delete,"** not "Alt + Delete." These notes use the correct shortcut.

SHORT QUESTIONS — CHAPTER 3

1. What is Search engine? Discuss any two. ***[Exercise]***
A search engine is a website used to look up information on the internet. Two well-known ones are Google and Yahoo.

2. Differentiate between 2D and 3D Shapes. ***[Exercise]***
A 2D shape has only length and width, like a flat square on paper. A 3D shape adds a third dimension, height, which gives it depth and makes it look solid.

3. Define software. ***[Exercise]***
A set of instructions, data and programs used to run a computer and carry out a given task is called software.

4. Differentiate between utility software and device drivers. ***[Exercise]***
A utility program carries out a management task, such as scanning for viruses or cleaning disk space. A device driver instead lets one specific hardware part, such as a printer, talk to the computer.

5. Enlist Language Translators. ***[Exercise]***
Four language translators: Assembler, Compiler, Interpreter, and a Language Translator program in general converts code for the machine.

6. Write a note on Snipping tools in windows. ***[Exercise]***
The Snipping Tool captures a picture of any chosen part of the screen; on Windows 10 and 11 it opens with Windows + Shift + S, and it offers free-form, rectangular, window, and full-screen capture modes.

7. Write down the usage of Notification area in windows. ***[Exercise]***
Sitting on the right side of the taskbar, the notification area shows the current date and time along with quick icons for things like battery, Wi-Fi, volume and the action center.

8. Differentiate between file and folder. ***[Exercise]***
A file holds one piece of stored content, such as a document or a photo. A folder is a container used to group several files together so they stay organized.

9. What is productivity software? ***[Exercise]***
Software used to produce a text document, spreadsheet, presentation, or piece of digital art is called productivity software.

10. Discuss Educational software. ***[Exercise]***
Educational software supports learning and research work; dictionaries, encyclopedias, educational games and mapping tools like Google Maps all fall under this category.

11. What are device drivers? Give an example. ***[Book concept]***
Device drivers let computer hardware parts communicate with the computer; a printer driver is one example.

12. What is a shortcut? ***[Book concept]***
A shortcut is a clickable link placed on the desktop or in a folder that takes the user straight to the original file.

13. How can a file be restored after deletion? ***[Book concept]***
Open the recycle bin, right-click the file that needs to come back, and choose the "Restore" option to send it to its original location.

14. Name the three window controls on the title bar. ***[Book concept]***
Close, Maximize and Minimize are the three buttons found on the top right of an active window's title bar.

15. Where should the fingers rest for typing? ***[Book concept]***
The left-hand fingers rest on A, S, D and F, while the right-hand fingers rest on J, K, L and the semicolon key; this is called the home row.

MULTIPLE CHOICE QUESTIONS — CHAPTER 3

1. The window in use on the desktop is known as a _____. ***[Exercise MCQ 1]***

a) Actual b) Dynamic **c) Active** d) Live

2. The process of moving from one webpage to another is known as: ***[Exercise MCQ 2]***

a) Changing b) Switching c) Transferring **d) Browsing**

3. Antivirus is an example of: ***[Exercise MCQ 3]***

a) Application software **b) Utility program** c) Device driver d) Operating system

4. A software used by teachers and students for enhancing learning skills and activities is an example of: ***[Exercise MCQ 4]***

a) System software **b) Educational software** c) Gaming software d) Operating system

5. A software developed to do a specific task on the computer is called: ***[Exercise MCQ 5]***

a) Operating system b) Entertainment software c) Gaming software **d) Application software**

6. Which of the following tool is in Paint 3D and is used to copy a shape? ***[Exercise MCQ 6]***

a) Crop tool b) Snipping tool **c) Clone stamp tool** d) Curve tool

7. We can change the size of the marker using the _____. ***[Exercise MCQ 7]***

a) Thickness slider b) Eraser c) Canvas d) Crayon

8. _____ is the link to the original file. ***[Exercise MCQ 8]***

a) Icon **b) Shortcut** c) Taskbar d) Folder

9. _____ is the collection of files. ***[Exercise MCQ 9]***

a) Folder b) Desktop c) Icons d) Applications

10. _____ is a website that allows us to search for information from the internet. ***[Exercise MCQ 10]***

a) Web browser b) Website c) Applications **d) Search engine**

11. Which of the following is an example of system software?

a) Operating system b) Microsoft Word c) a video game d) a web browser

12. A device driver is needed so that:

a) a document can be printed **b) hardware can communicate with the computer** c) files can be compressed d) a webpage can load

13. Mac OS was first introduced for:

a) Android phones b) Linux servers **c) Macintosh personal computers** d) embedded systems

14. Android is based upon:

a) Windows Kernel **b) Linux Kernel** c) Mac Kernel d) none of these

15. Loading the operating system into main storage when the computer starts is called:

✓ **a) Booting process** | b) File management | c) Device management | d) Security management

16. The OS function that controls peripherals such as printers is called:

a) Memory management | ✓ **b) Device management** | c) Application management | d) Booting process

17. The small bar at the bottom of the desktop is called the:

a) Background | ✓ **b) Taskbar** | c) Notification area | d) Start screen

18. The keyboard shortcut used to open the Snipping Tool on Windows 10/11 is:

a) Ctrl + Shift + S | ✓ **b) Windows + Shift + S** | c) Alt + Shift + S | d) Windows + Alt + S

19. Which title bar button hides a window and places it on the taskbar?

a) Close | b) Maximize | ✓ **c) Minimize** | d) Restore

20. A file can store:

a) only text | b) only photos | ✓ **c) text, photo, audio, video, document or presentation** | d) only a program

21. After deleting a file, it first goes to the:

a) Taskbar | ✓ **b) Recycle bin** | c) Clipboard | d) Notification area

22. The correct Windows shortcut to permanently delete a file (skipping the recycle bin) is:

a) Alt + Delete | ✓ **b) Shift + Delete** | c) Ctrl + Delete | d) Tab + Delete

23. Paint 3D is best described as a refresh of:

a) Microsoft Word | ✓ **b) Microsoft Paint** | c) Microsoft Excel | d) Windows Store

24. In geometry, 3D shapes are defined by:

a) length and width only | ✓ **b) length, width and height** | c) colour only | d) area only

25. An aimless or random drawing made with the 3D shapes tool is called a:

a) sticker | b) stamp | ✓ **c) doodle** | d) canvas

26. A web browser is used to:

a) type documents | ✓ **b) open webpages and view internet information** | c) edit images | d) play games only

27. Which of the following is NOT an example of a web browser?

a) Google Chrome | b) Mozilla Firefox | ✓ **c) Google** | d) Safari

28. Typing on the keyboard using the correct finger for each key is called:

a) Browsing | ✓ **b) Keyboarding** | c) Booting | d) Snipping

29. While keyboarding, the elbow should be bent at an angle of:

a) 45 degrees | ✓ **b) 90 degrees** | c) 120 degrees | d) 180 degrees

30. On the home row, the left-hand fingers rest on:

a) J, K, L | ✓ **b) A, S, D, F** | c) Q, W, E, R | d) Z, X, C, V